

3D Coordinates Extension 2 — English Worksheet (Intermediate)

Topic: 3D Mob Heads (x, y, z) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Lesson 3) · **Time:** about 40 minutes

Build a **creeper head** in 3D. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict 🧠

Read each set. Write a sentence for each.

```
place green block at (3, 5, 1)
place green block at (3, 5, 2)
place green block at (3, 5, 3)
```

Only z changes. What happens to the line?


```
place green block at (4, 5, 1)
place green block at (4, 6, 1)
place green block at (4, 7, 1)
```

Only y changes. What happens to the line?


```
place green block at (3, 5, 1)
place black block at (3, 5, 1)
```

Same (x, y, z). What do you see, and why?

2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

Bug A — This block should sit one step **deeper**, at (3, 5, 2). It only has two numbers, so it lands on the front.

```
place green block at (3, 5)
```

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences)

Bug B — The eye should be **up high and one step deep**: (4, 7, 2). The last two numbers are swapped.

```
place black block at (4, 2, 7)
```

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences)

3 · Show Your Work 📷 🎥

Look at the 🟢 creeper head in the world. Walk around it. Then build **one full head** on your home spot, with a face.

Front face (keep $z = 1$): fill the green square, then add the creeper face — black eyes and a black mouth on the front.

Add depth: build the same front face again at $z = 2$, then $z = 3$, until your head is as deep as the real one.

Face on front only: at $z = 2$ and $z = 3$, place plain green where the eyes and mouth were.

💡 Tip: the eyes and mouth only show on the **front**. The sides and back are plain green.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.